Macro-F1 Is Not Enough: Calibration of Behavior Classifiers on Imbalanced Learning Logs

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Abstract

Behavior classifiers in learning analytics are usually summarized by macro-F1. We show on three datasets that models with similar macro-F1 can differ widely in calibration on minority states, which matters when predictions trigger interventions. We recommend reporting per-class F1, expected calibration error per class, and a reliability diagram for the state that will drive an intervention.

1 Introduction

An intervention fires when a model's probability of "stuck" crosses a threshold. Macro-F1 says nothing about whether a probability of 0.8 means stuck 80% of the time. On imbalanced data, the minority state that triggers the intervention is the one most likely to be miscalibrated.

2 Experiments

We trained four model families on three behavior datasets (minority-state prevalence 6% to 18%) and measured macro-F1 and expected calibration error (ECE, 15 bins) for the minority state.

Model	Macro-F1	Minority F1	Minority ECE
Logistic regression	0.70	0.52	0.04
Gradient boosting	0.74	0.55	0.11
Gradient boosting + class weights	0.74	0.61	0.19
GRU	0.73	0.57	0.14

Class weighting raised minority F1 but nearly doubled its calibration error: the model learned to over-predict the minority state. Temperature scaling on a grouped validation split reduced ECE to 0.05 without changing F1.

3 Reliability diagrams

Reliability diagrams make the problem visible. For the class-weighted gradient-boosting model, windows predicted "stuck" with probability 0.7-0.8 were actually stuck 41% of the time. An instructor dashboard that flags students above 0.7 would send a TA to the wrong student more often than the right one, even though the model's macro-F1 matches the best model in the table.

Predicted probability bin	Windows	Observed rate (weighted GB)	Observed rate (after temperature scaling)
0.5-0.6	412	0.29	0.53
0.6-0.7	288	0.34	0.62
0.7-0.8	197	0.41	0.73
0.8-0.9	121	0.55	0.84

0.9-1.0	64	0.70	0.91

4 Splits matter for calibration too

Calibration measured on a random split looked better than on a split grouped by student (ECE 0.07 vs 0.11 for the unweighted model). A model that has seen a student's other windows is more confident about that student, and calibrated on that confidence. Temperature should be fit on held-out students.

5 A worked example

Consider an intervention that pauses the assistant and suggests a TA check-in when P(stuck) > 0.8. With the weighted, uncalibrated model, 45% of check-ins would target students who were not stuck. After calibration on a grouped validation split, the false share drops to 16% while the number of stuck students reached falls only slightly (from 118 to 109 of 152 in our test set). The intervention cost depends on calibration far more than on the second decimal of macro-F1.

6 Recommendations

- Report per-class F1 alongside macro-F1.
- Report ECE for any class used to trigger an intervention, measured on a split grouped by student.
- Calibrate after class weighting.
- Publish the reliability diagram for the intervention class.

7 Limitations

ECE is sensitive to binning; we report results for 10 and 20 bins in the supplement. Our datasets come from two platforms and may not reflect the class balance of other courses.

References

[1] Ferreira, A. and Kowalczyk, P. Group-aware splits for educational sequence data. JEDM 15, 2 (2023), 44-67.